A Generalization of Macdonald functions and Kontorovich–Lebedev transform using generalized hyperbolic functions

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Abstract

The Kontorovich–Lebedev transform is a powerful index transform in mathematics and theoretical physics, built upon modified Bessel functions of the second kind. In this paper, we introduce a novel generalization of both the modified Bessel functions and their corresponding Kontorovich–Lebedev transform framework, utilizing Riccati's generalized hyperbolic functions as formalized by Ungar [2]. We prove orthogonality of the generalized kernel via distribution theory, derive the corresponding inversion formula, establish asymptotic behavior at both boundaries, derive a recurrence relation for even order, and prove Parseval's identity for the extended transform.

Contents

1 Prerequisite

Define Generalized $\cosh(x)$ as the "main cyclic function" of the n-th order as[2]:

$$\cosh_n(x) := \sum_{k=0}^{\infty} \frac{x^{nk}}{(nk)!} = \frac{1}{n} \sum_{m=0}^{n-1} e^{\omega^m x}, \quad \{x \in \mathbb{R}, n \in \mathbb{N}, \omega = e^{\frac{2\pi i}{n}}\}$$

Which is positive for all real valued x in $e^{\omega x}$
Define the generalization of $\sinh(x)$ as the family of functions $\mathrm{sinh_n\,K}(x)$ where

$$\mathrm{sinh_n\,K}(x) := \frac{d^K}{dx^K} \cosh_n(x) \implies \mathrm{sinh_n\,K}(x) = \frac{1}{n} \sum_{m=0}^{n-1} \omega^{mK} e^{\omega^m x}$$

Let $K_\nu(x)$ denote the Modified Bessel function of the second kind using integral representation as [1]

$$K_\nu(x) = \int_0^\infty e^{-x\cosh(t)} \cosh(\nu t) dt \quad \mathrm{Re}(x) > 0$$

Let $\mathcal{F}_c\{f(t)\}(x) = \int_0^\infty f(t)\cos(xt)\,dt$ denote the Fourier cosine transform of $f(t)$

2 Motivation and Generalization

when defining $\mathrm{sech_n}(t)$ using $\cosh_n(t)$ we write it as $\mathrm{sech_n}(t) = \frac{1}{\cosh_n(t)}$, Observe that using $\frac{1}{A} = \int_0^\infty e^{-tA} dt$ we can write $\mathrm{sech_n}(x)$ as follows

$$\mathrm{sech_n}(t) = \int_0^\infty e^{-x\cosh_n(t)} dx$$

Taking the Fourier cosine transform of such a function returns

$$\mathcal{F}_c\{\mathrm{sech_n}(t)\}(\nu) = \int_0^\infty \left[\int_0^\infty e^{-x\cosh_n(t)} dx\right] \cos(\nu t)\,dt$$

Swapping the order of integration via Fubini's theorem then yields:

$$= \int_0^\infty \left[\int_0^\infty e^{-x\cosh_n(t)} \cos(\nu t)\,dt\right] dx = \int_0^\infty K_{i\nu}^{(n)}(x)\,dx$$

At $n = 2$ we see the definition of $K_{i\nu}(x)$, thus a generalization can be considered

$$K_{i\nu}^{(n)}(x) = \int_0^\infty e^{-x\cosh_n(t)} \cos(\nu t) dt \implies K_\nu^{(n)}(x) = \int_0^\infty e^{-x\cosh_n(t)} \cosh(\nu t) dt$$

3 Singular behavior of $K_0^{(n)}(x)$

The integral of $K_0^{(n)}(x)$ is

$$K_0^{(n)}(x) = \int_0^\infty e^{-x\cosh_n(t)}dt$$

Let's split this integral into two integrals

$$K_0^{(n)}(x) = \int_0^A e^{-x\cosh_n(t)}dt + \int_A^\infty e^{-x\cosh_n(t)}dt$$

as $x \to 0$ the first term equal A, and for the second term we know that the power grows like $\frac{1}{n}e^t$ for large t,now we can do a u-substitution $du = \frac{x}{n}e^t dt = u\,dt$, so $dt = \frac{du}{u}$, when $t = A$, $u = \frac{x}{n}e^A$. When $t \to \infty$, $u \to \infty$

$$I = \int_A^\infty e^{-\frac{x}{n}e^t}dt \approx I_{\text{tail}} \approx \int_{\frac{x}{n}e^A}^\infty \frac{e^{-u}}{u}du$$

This is the definition of the Exponential integral $E_1(x) = \int_1^\infty \frac{e^{-xt}}{t}\,dt$, and as $x \to 0$ in this integral we get$E_1(x) = -\gamma - \ln(x) + x - \dots$ where γ is the Euler-Mascheroni constant, Substituting $x = \frac{x}{n}e^A$

$$I \approx -\ln\left(\frac{xe^A}{n}\right) - \gamma = -\ln(x) + \ln(n) - A - \gamma$$

which means that as $x \to 0$, the constant terms become irrelevant compared to the term that goes to infinity, Therefore $K_0^{(n)}(x) \sim -\ln(x)$

4 Boundary Behavior of $K_{i\mu}^{(n)}(x)$

outside of the $x = 0$ case , we can find the behavior of the function using $\cosh_n(x) = \sum_{k=0}^\infty \frac{x^{nk}}{(nk)!} = 1 + \frac{t^n}{n!} + \mathcal{O}(t^{2n})$ We can apply Laplace's method to find the exact asymptotic behavior of the kernel as $x \to \infty$:

$$K_{i\mu}^{(n)}(x) = \int_0^\infty e^{-x\cosh_n(t)}\cos(\mu t)\,dt \sim e^{-x}\int_0^\infty e^{-x\frac{t^n}{n!}}\,dt$$

By making the substitution $u = \frac{x}{n!}t^n$, the integral evaluates directly to a Gamma function

$$K_{i\mu}^{(n)}(x) \sim e^{-x}x^{-1/n}\,(n!)^{1/n}\,\Gamma\left(1 + \frac{1}{n}\right) \quad \text{as } x \to \infty$$

at $n = 2$ $e^{-x}x^{-1/2}\,(n!)^{1/2}\,\Gamma\left(\frac{3}{2}\right) = \sqrt{\frac{\pi}{2x}}e^{-x}$ which is the classic result for $K_\nu(x)$ and as $x \to 0^+$, the tail of the integral is dominated by the largest real exponential branch of $\cosh_n(t) \sim \frac{1}{n}e^t$

$$K_{i\mu}^{(n)}(x) = \begin{cases} \mathcal{O}(\log(1/x)) & \text{if } \mu = 0 \\ \mathcal{O}(1) \text{ (bounded oscillation)} & \text{if } \mu > 0 \end{cases} \quad \text{as } x \to 0^+$$

5 Proofs for Generalized Kontorovich–Lebedev transform

5.1 Orthogonality of $K_{i\beta}^{(n)}$ proof using distribution theory

Since $K_{iv}(x)$ is orthogonal with respect to the measure $\frac{dx}{x}$,let's try to prove it with respect to the measure $w(x)dx$
let

$$\int_0^\infty K_{i\nu}^{(n)}(x)K_{iv}^{(n)}(x)w(x)dx = \Omega(n,\mu,\nu)\delta(\mu-\nu)$$

where $\delta(\mu-\nu)$ is the Dirac delta function, $\Omega(n,\mu,\nu)$ is an unknown normalization factor depending on n,μ,ν
Expand using the left-hand side integral definitions

$$I = \int_0^\infty \int_0^\infty \cos(\mu t)\cos(\nu u)\int_0^\infty w(x)e^{-x(\cosh_n(t)+\cosh_n(u))}dxdudt$$

assume $w(x)=\frac{1}{x}$ as in the case of $n=2$, then integrate by parts $\frac{\partial}{\partial t}$ then $\frac{\partial}{\partial u}$.

$$I = \frac{1}{\mu\nu}\int_0^\infty \int_0^\infty \sin(\mu t)\sin(\nu u)\frac{\mathrm{sinh_n}(t)\,\mathrm{sinh_n}(u)}{(\cosh_n(t)+\cosh_n(u))^2}dudt$$

for the result to be a delta function, the kernel $\frac{\mathrm{sinh_n}(t)\,\mathrm{sinh_n}(u)}{(\cosh_n(t)+\cosh_n(u))^2}$must act like a sharp ridge along the line $t=u$ as $t,u\to\infty$. From the definition of $\cosh_n(x)$ the $\lim_{x\to\infty}\cosh_n(x)\approx\frac{1}{n}e^x$, We can substitute these approximations into the kernel

$$\frac{\mathrm{sinh_n}(t)\,\mathrm{sinh_n}(u)}{(\cosh_n(t)+\cosh_n(u))^2} \approx \frac{(\frac{1}{n}e^t)(\frac{1}{n}{}^u)}{(\frac{1}{n}e^t+\frac{1}{n}e^u)^2} = \frac{e^{t+u}}{(e^t+e^u)^2} = \frac{e^{t+u}}{(2e^{(t+u)/2}\cosh(\frac{t-u}{2}))^2}$$

$$\frac{e^{t+u}}{4e^{(t+u)/2}\cosh(\frac{t-u}{2})^2} = \frac{1}{4\cosh(\frac{t-u}{2})^2}$$

This asymptotic approximation is valid in the limit $t,u\to\infty$ where the integral is dominated, as the kernel exhibits a sharp peak along $t=u$. Contributions from finite t,u are negligible in forming the delta function.
let $y=\frac{t-u}{2}$ and $x=\frac{t+u}{2}$, therefore $t=x+y, u=x-y$ and $dudt=2dxdy$, then we use the $\sin(a)\sin(b)$ formula

$$I = \frac{1}{\mu\nu}\int_{-\infty}^\infty \frac{1}{4\cosh(y)^2}\int_{|y|}^\infty [\cos((\mu-\nu)x+(\mu+\nu)y)-\cos((\mu+\nu)x+(\mu-\nu)y)]dxdy$$

In distribution theory, $\int_0^\infty \cos(Ax+B)dx = \pi\delta(A)\cos(B)$
When $A=\mu-\nu$, this formula returns, otherwise it vanishes
when $\delta(\mu-\nu)$ doesn't vanish(at $\mu=\nu$), the integral is

$$I = \frac{\pi\delta(\mu-\nu)}{4\mu^2}\int_{-\infty}^\infty \frac{\cos(2\mu y)}{\cosh(y)^2}dy$$

$$I = \frac{\pi\delta(\mu-\nu)}{4\mu^2} \times \frac{2\pi\mu}{\sinh(\pi\mu)} = \frac{\pi^2}{2\mu\sinh(\pi\mu)}\delta(\mu-\nu)$$

Which proves its orthogonality, in fact, its orthogonality is invariant under n order

5.2 Proof of generalized Kontorovich–Lebedev using Fourier cosine

The Kontorovich–Lebedev transform for a function $f(x)$ is defined as[3]

$$F(\nu) = \int_0^\infty f(x)K_{i\nu}(x)\frac{dx}{x}$$

with the inverse being

$$f(x) = \frac{2}{\pi^2}\int_0^\infty F(\nu)\nu\sinh(\pi\nu)K_{i\nu}(x)d\nu$$

Similarly , let the generalized Kontorovich-Lebedev transform be defined as

$$F_n(\mu) = \int_0^\infty f(x)K_{i\mu}^{(n)}(x)\frac{dx}{x}$$

We propose that the inverse transform is given by the synthesis operator $g(x)$:

$$g(x) = \frac{2}{\pi^2}\int_0^\infty F_n(\nu)\nu\sinh(\pi\nu)K_{i\nu}^{(n)}(x)d\nu$$

To prove that $g(x) = f(x)$, we project $g(x)$ back into the transform space by multiplying both sides by $K_{i\mu}^{(n)}(x)/x$ and integrating over $x \in (0,\infty)$

$$\int_0^\infty g(x)K_{i\mu}^{(n)}(x)\frac{dx}{x} = \int_0^\infty \left[\frac{2}{\pi^2}\int_0^\infty F_n(\nu)\nu\sinh(\pi\nu)K_{i\nu}^{(n)}(x)d\nu\right]K_{i\mu}^{(n)}(x)\frac{dx}{x}$$

Assuming $f(x)$ belongs to a suitable weighted functional space such that Fubini's theorem applies, we exchange the order of integration.

$$= \frac{2}{\pi^2}\int_0^\infty F_n(\nu)\nu\sinh(\pi\nu)\left[\int_0^\infty K_{i\nu}^{(n)}(x)K_{i\mu}^{(n)}(x)\frac{dx}{x}\right]d\nu$$

Substitute the orthogonality relation

$$= \frac{2}{\pi^2}\int_0^\infty F_n(\nu)\nu\sinh(\pi\nu)\left[\frac{\pi^2}{2\nu\sinh(\pi\nu)}\delta(\nu-\mu)\right]d\nu$$

$$= \int_0^\infty F_n(\nu)\delta(\nu-\mu)d\nu = F_n(\mu)$$

Because the transform of $g(x)$ is identical to the transform of $f(x)$, and assuming the family of kernels $K_{i\nu}^{(n)}(x)$ is complete over the weighted space(n=2 this is actually the case , But it's still an open problem for $n > 2$), it follows that $g(x) = f(x)$ almost everywhere. This establishes the inversion formula invariantly for any order $n \geq 2$.

6 Recurrence Formula for $K_\beta^{(n)}(x)$

We start with integration by parts, let $u = e^{-x\cosh_n(t)}$ do $\frac{d}{dt}u = -x\sinh_n(t)e^{-x\cosh_n(t)}$, let $\frac{d}{dv} = \cosh(\beta t)$ so $v = \frac{1}{\beta}\sinh(\beta t)$

$$K_\beta^{(n)}(x) = \left[e^{-x\cosh_n(t)}\frac{\sinh(\beta t)}{\beta}\right]_0^\infty - \left(-\frac{x}{\beta}\int_0^\infty \sinh_n(t)e^{-x\cosh_n(t)}\sinh(\beta t)dt\right)$$

$$= \frac{x}{\beta}\int_0^\infty e^{-x\cosh_n(t)}\sinh_n(t)\sinh(\beta t)dt$$

we use the definition of $\sinh_n(t) = \frac{1}{n}\sum_{k=0}^{n-1}\omega^k e^{\omega^k t}$ and the euler form of $\sinh(\beta t) = \frac{e^{\beta t}-e^{-\beta t}}{2}$

$$K_\beta^{(n)}(x) = \frac{x}{2\beta}\int_0^\infty e^{-x\cosh_n(t)}\sum_{k=0}^{n-1}\omega^k\left[e^{(\omega^k+\beta)t} - e^{(\omega^k-\beta)t}\right]dt$$

$$= \frac{x}{2\beta}\sum_{k=0}^{n-1}\omega^k\int_0^\infty e^{-x\cosh_n(t)}e^{(\omega^k+\beta)t}dt - \int_0^\infty e^{-x\cosh_n(t)}e^{(\omega^k-\beta)t}dt$$

if we expand $e^{(\omega^k+\beta)t}$ and $e^{(\omega^k-\beta)t}$ using hyperbolic functions we get two kernels, one of them is $K_{\beta+\omega^k}^{(n)}(x), K_{-\beta+\omega^k}^{(n)}(x)$ and the other uses sinh instead of cosh, let's denote it with ${}^S K_{\beta+\omega^k}^{(n)}(x)$ and ${}^S K_{-\beta+\omega^k}^{(n)}(x)$
Because cosh is an even function and sinh is an odd function, we know that:$K_{-\beta+\omega^k}^{(n)} = K_{\beta-\omega^k}^{(n)}, {}^S K_{-\beta+\omega^k}^{(n)}(x) = -{}^S K_{\beta-\omega^k}^{(n)}(x)$
therefore

$$\frac{2n\beta}{x}K_\beta^{(n)}(x) = \sum_{k=0}^{n-1}\omega^k\left(\left[K_{\beta+\omega^k}^{(n)}(x) - K_{\beta-\omega^k}^{(n)}(x)\right] + \left[{}^S K_{\beta+\omega^k}^{(n)}(x) + {}^S K_{-\beta+\omega^k}^{(n)}(x)\right]\right)$$

when n is even, we can group the sum by ω^k and $-\omega^k$ since the root, and its conjugate must appear, since the $K_{\beta+\omega^k}^{(n)}(x)$ function is even it won't have any problems, but for the other function ${}^S K_{\beta+\omega^k}^{(n)}(x)$, it will cancel itself due to it being an odd function, we will be left out with $2\times$ the sum of $K_{\beta+\omega^k}^{(n)}(x)$ functions, and dividing both sides we get

$$\frac{n\beta}{x}K_\beta^{(n)}(x) = \sum_{k=0}^{\frac{n}{2}-1}\omega^k\left(\left[K_{\beta+\omega^k}^{(n)}(x) - K_{\beta-\omega^k}^{(n)}(x)\right]\right)$$

which is the standard case when $n = 2$ which means that $\omega = -1$ and $K_\beta(x) = K_\beta(x)$

$$\frac{2\beta}{x}K_\beta(x) = \sum_{k=0}^{\frac{2}{2}-1}(-1)^k\left(\left[K_{\beta+(-1)^k}(x) - K_{\beta-(-1)^k}(x)\right]\right) = K_{\beta+1}(x) - K_{\beta-1}(x)$$

But this clean form doesn't hold for odd n

7 Parseval's identity for generalized Kontorovich–Lebedev transform

from the orthogonality, we can follow
Let $F_n(\nu)$ and $H_n(\nu)$ be the generalized KL transforms of $f(x)$ and $h(x)$ respectively

$$\int_0^\infty f(x)h(x)\frac{dx}{x} = \int_0^\infty f(x)\left[\frac{2}{\pi^2}\int_0^\infty H_n(\nu)\nu\sinh(\pi\nu)K_{i\nu}^{(n)}(x)d\nu\right]\frac{dx}{x}$$

Swap the integrals:

$$= \frac{2}{\pi^2}\int_0^\infty H_n(\nu)\nu\sinh(\pi\nu)\left[\int_0^\infty f(x)K_{i\nu}^{(n)}(x)\frac{dx}{x}\right]d\nu$$

$$\int_0^\infty f(x)h(x)\frac{dx}{x} = \frac{2}{\pi^2}\int_0^\infty F_n(\nu)H_n(\nu)\nu\sinh(\pi\nu)d\nu$$

therefore

$$\int_0^\infty f(x)^2\frac{dx}{x} = \frac{2}{\pi^2}\int_0^\infty F_n(\nu)^2\nu\sinh(\pi\nu)d\nu$$

8 Functional Framework and Convergence Requirements

define the underlying functional spaces for the test functions $f(x)$. Let $L_2((0,\infty); x^{-1}dx)$ denote the weighted Lebesgue space of square-integrable functions under the invariant measure dx/x, equipped with the norm

$$\|f\| = \left(\int_0^\infty |f(x)|^2\frac{dx}{x}\right)^{1/2} < \infty \tag{1}$$

For pointwise and absolute convergence of the transform integral, the behavior of the kernel $K_{i\mu}^{(n)}(x)$ necessitates that $f(x)$ satisfies the following structural bounds:

1. **Local condition at the origin:** $f(x) = \mathcal{O}(x^\alpha)$ as $x \to 0^+$ for some $\alpha > 0$, ensuring integrability against the logarithmic singularity of the kernel and the x^{-1} weight.

2. **Growth condition at infinity:** $f(x) = \mathcal{O}(e^{cx})$ as $x \to \infty$ for some $c < 1$, which is strictly dominated by the asymptotic decay of the generalized kernel, $K_{i\mu}^{(n)}(x) \sim \mathcal{O}(e^{-x}x^{-1/n})$.

9 Table of some transformed functions

Small Table for specific transformed functions

Object Function $f(x)$	**Generalized Transform** $F_n(\nu)$
$x_0\delta(x-x_0), \quad x_0>0$	$K_{i\nu}^{(n)}(x_0)$
$xe^{-ax}, \quad a \geq 0$	$\displaystyle\int_0^\infty \frac{\cos(\nu t)}{a+\cosh_n(t)}\,dt$
$x^\alpha, \quad \alpha>0$	$\displaystyle\Gamma(\alpha)\int_0^\infty \cos(\nu t)\,[\mathrm{sech}_n(t)]^\alpha\,dt$

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